

- 7 The bacterium, *Escherichia coli*, can use glucose or disaccharides, such as lactose, in its metabolism. Lactose needs to be hydrolysed by the enzyme β -galactosidase to form glucose and galactose, which can then be used by *E. coli*.

The production of β -galactosidase is controlled by a length of DNA called the *lac* operon.

- (a) Fig. 7.1 shows the *lac* operon when lactose is **absent**.

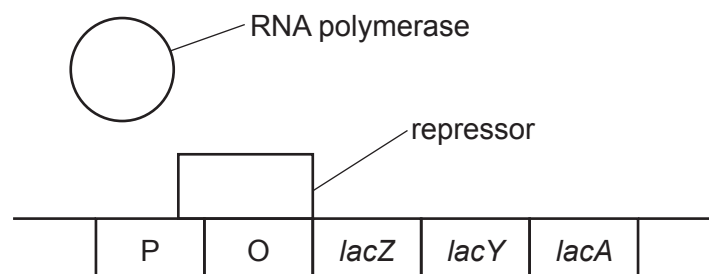


Fig. 7.1

On Fig. 7.2, draw the positions of RNA polymerase and the repressor molecule when lactose is **present**.

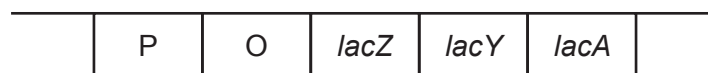


Fig. 7.2

[2]

- (b) The protein coded for by *lacY* is not involved in the control of gene expression.

- (i) Name the type of gene represented by *lacY*.

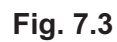
..... [1]

- (ii) Name the protein product coded for by *lacY* **and** state the precise role of this protein.

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 [2]

- Fig. 7.3 shows the population growth curve obtained for this investigation.



[4]

[Total: 9]